

Swap x_9 with x_7

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$

$$\Rightarrow x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 10 - 12x_2 - 7.2x_3 - 0.47x_1 - 2.7x_5 - 0.17x_6 - 1.6x_9$$

$$x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

$$\Rightarrow x_4 = 2.1 - 0.83x_2 + 0.17x_3 - 0.13x_1 - 0.7x_5 - 0.83x_6 - 0.23x_9$$

$$x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7$$

$$\Rightarrow x_8 = 8.6 + 2x_2 - 8x_3 + 0.2x_1 - 18x_5 - 9x_6 - 1.4x_9$$